Perception Over Adoption: Perceived AI Job Threat Predicts Developer Job Satisfaction More Strongly than AI Tool Use

Evidence from the 2024 Stack Overflow Developer Survey

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Abstract

The integration of artificial intelligence tools into software development has prompted claims that AI adoption drives developer productivity and well-being. Using data from the 2024 Stack Overflow Developer Survey ($N = 17{,}696$), this study tests whether perceived AI job threat or actual AI tool adoption is a stronger predictor of developer job satisfaction. Ordinary least-squares regression and group-comparison analyses reveal that developers who perceive AI as a threat to their jobs report significantly lower job satisfaction than those who do not ($d = 0.33$, $\eta^2 = 0.012$, $p < 10^{-47}$), with a mean difference of 0.67 points on a 0–10 scale. By contrast, whether developers currently use AI tools is effectively unrelated to job satisfaction ($d = 0.031$, $p = 0.14$). The AIThreat effect persists after controlling for experience, remote-work status, and organisation size ($\beta = -0.65$, $p < 10^{-44}$), and holds equally among current AI tool users. These results suggest that psychological threat perception, not adoption behaviour, is the mechanism linking AI to developer job satisfaction.

1 Introduction

Artificial intelligence coding assistants have moved from novelty to daily infrastructure in professional software development. Industry surveys consistently report that a large majority of developers have experimented with or adopted tools such as GitHub Copilot, ChatGPT, or similar assistants [Vaillant et al., 2024]. The dominant narrative accompanying this shift is that AI adoption increases productivity and, by extension, developer well-being: more capable tools should translate into more satisfying work.

Yet adoption rates and subjective well-being need not move together. A growing body of research suggests that the *psychological response* to AI—rather than its practical use—may be the more consequential variable. Sadeghi [2024] finds that AI deployment in organisational settings simultaneously raises concerns about employment stability, fairness, and privacy, with the net effect on worker contentment dependent on implementation context rather than on whether employees use the technology. At the societal level, Giuntella et al. [2025] show, using longitudinal German panel data, that workers in highly AI-exposed occupations do not report lower life satisfaction or greater job insecurity than comparable workers in less-exposed roles—suggesting that aggregate exposure by itself is not harmful. For individual well-being, however, the picture appears different: Feng et al. [2025] demonstrate that AI-related workplace anxiety directly reduces life satisfaction through heightened negative affect, with social support acting as a buffer.

In a systematic review of 92 workplace AI studies, Soulami et al. [2024] identify *workplace autonomy* and *psychological health* as the two most consistently observed mediating themes: AI adoption affects employees primarily through changes in perceived control and stress, not through raw exposure or tool usage per se. These findings collectively point toward a hypothesis that has received little direct empirical testing in the software development domain: that *perceived AI threat*, rather than *AI adoption status*, is the dominant predictor of developer job satisfaction.

The present study tests this hypothesis using data from the 2024 Stack Overflow Developer Survey, one of the largest annual samples of software professionals worldwide. Specifically, we ask:

> *Does perceived AI job threat predict lower developer job satisfaction, and does this effect exceed the effect of actual AI tool adoption?*

Our null hypothesis (H_0) is that AIThreat does not predict job satisfaction after controlling for AI tool adoption and standard covariates ($\beta = 0$). The alternative (H_1) is that AIThreat is a significant negative predictor ($\beta < 0$, $d \geq 0.25$) and that its effect substantially exceeds that of adoption behaviour.

2 Methodology

2.1 Data Source

Data are from the *2024 Stack Overflow Annual Developer Survey*, a publicly available cross-sectional survey of software developers worldwide. The full dataset contains 65,437 responses. The primary outcome variable is JobSat (0–10, integer), answered by 29,126 respondents (55.5 % item non-response). Exploratory analyses indicate that non-response to JobSat is concentrated among hobby coders and students (i.e. respondents without a professional employment context), suggesting a Missing At Random (MAR) pattern conditional on MainBranch.

2.2 Analytic Sample

The analytic sample was restricted to respondents who (a) provided a JobSat value, (b) gave a binary response to AIThreat ("Yes" or "No"; "I'm not sure" and "NA" excluded from primary analysis), (c) reported a current or near-term AISelect status ("Yes" or "No, but I plan to soon"), and (d) had valid values for all covariates (RemoteWork, OrgSize). This yielded $N = 17{,}696$.

2.3 Variables and Encoding

Outcome. JobSat: continuous 0–10 (higher = more satisfied).

Primary predictor. AIThreat_Yes: 1 if the respondent views AI as a threat to their job, 0 otherwise (reference: No).

Secondary predictor. AISelect_Yes: 1 if the respondent currently uses AI tools, 0 if they plan to use them soon (reference). Note: within the AIThreat Yes/No subsample, the category "No, and I don't plan to" is absent, so the reference becomes "plans to use soon."

Covariates. YearsCodePro (continuous; 21.1 % missing, imputed with sample median of 8 years under the MAR assumption), RemoteWork (three dummies; reference: In-person), OrgSize (eight dummies; reference: Freelancer / just me).

2.4 Statistical Analysis

We conducted (1) Welch t-tests and one-way ANOVA with η^2 for bivariate comparisons; (2) OLS regression in two nested models—Model 1 (AIThreat + AISelect without covariates) and Model 2 (full model)—to assess β coefficients, R^2, and incremental ΔR^2; (3) Cohen's d for standardised effect sizes; (4) variance inflation factors (VIF) to assess multicollinearity; and (5) a robustness check restricted to current AI tool users (n = 14,538). Effect-size interpretation follows Cohen's conventions ($d \geq 0.20$ small, ≥ 0.50 medium; $f^2 \geq 0.02$ small, ≥ 0.15 large).

3 Results

3.1 Descriptive Statistics

Table 1 reports means and standard deviations for the two AIThreat groups and the two AISelect groups.

Table 1: Descriptive statistics for JobSat by AIThreat and AISelect.

Group	n	Mean	SD	Median
AIThreat				
AIThreat = No	15,414	7.099	2.025	7
AIThreat = Yes	2,282	6.426	2.285	7
AISelect				
Currently uses AI	14,538	7.024	2.089	7
Plans to use AI	3,158	6.960	1.994	7

The mean difference between AIThreat groups is 0.67 points on the 0–10 scale. The mean difference between AISelect groups is only 0.06 points.

3.2 Primary Analysis: AIThreat vs. AISelect

A Welch t-test confirmed a significant mean difference between AIThreat groups ($t = 13.32$, $p < 2.5 \times 10^{-39}$, Figure 1). One-way ANOVA yielded $F(1, 17{,}694) = 212.27$, $p < 8.3 \times 10^{-48}$, $\eta^2 = 0.012$. The standardised effect size was Cohen's $d = 0.33$ (small-to-moderate range), compared to $d = 0.031$ for AISelect (negligible).

Figure 2 visualises this contrast directly: AIThreat exceeds the small-effect threshold by a factor of 1.6, while AISelect falls far below it.

3.3 Regression Analysis

Table 2 presents the two OLS models.

In Model 1, AIThreat_Yes carries a coefficient of $\beta = -0.673$ ($p < 0.001$; 95 % CI: $[-0.763, -0.582]$), while AISelect_Yes is non-significant ($\beta = 0.060$, $p = 0.14$). After adding all covariates in Model 2, AIThreat_Yes remains highly significant ($\beta = -0.651$, $p < 10^{-44}$; 95 % CI: $[-0.741, -0.562]$), while AISelect_Yes becomes marginally significant but substantively trivial ($\beta = 0.096$, $p = 0.018$).

The incremental R^2 attributable to AIThreat above covariates and AISelect is $\Delta R^2 = 0.011$ ($f^2 = 0.011$), placing it in the small-effect range by Cohen's f^2 convention.

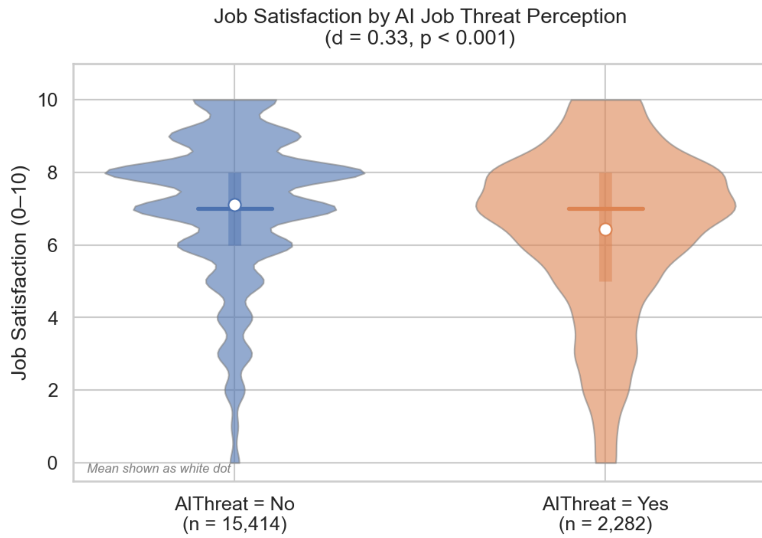

Figure 1: Distribution of JobSat by AIThreat group. Violin plot with embedded box (IQR); white dots denote group means. $d = 0.33$, $p < 10^{-47}$.

Table 2: OLS regression predicting JobSat. Reference categories: AIThreat=No, AISelect=Plans-to-use, RemoteWork=In-person, OrgSize=Freelancer.

	Model 1		Model 2	
Predictor	β	p	β	p
Intercept	7.050	<0.001	6.644	<0.001
AIThreat_Yes	−0.673	<0.001	−0.651	<0.001
AISelect_Yes	0.060	0.140	0.096	0.018
YearsCodePro	—	—	0.028	<0.001
RemoteWork_Hybrid	—	—	0.242	<0.001
RemoteWork_Remote	—	—	0.336	<0.001
OrgSize dummies	—	—	n.s. (mostly)	—
N	17,696		17,696	
R^2	0.012		0.033	

All VIFs in Model 2 are below 6 (max: AISelect_Yes, VIF = 5.53), indicating acceptable multicollinearity.

3.4 Interaction and Robustness

The interaction term AIThreat × AISelect was non-significant ($f^2 = 0.0008$), indicating that the AIThreat effect does not depend on whether respondents use AI tools. As Figure 3 shows, the gap between AIThreat=No and AIThreat=Yes is similarly sized among current AI tool users (7.11 vs. 6.43) and among those planning to adopt (7.02 vs. 6.34).

A robustness check restricted to current AI tool users ($n = 14{,}538$) yielded $d = 0.325$ ($t = 11.89$, $p < 1.1 \times 10^{-31}$), virtually identical to the full-sample estimate ($d = 0.327$).

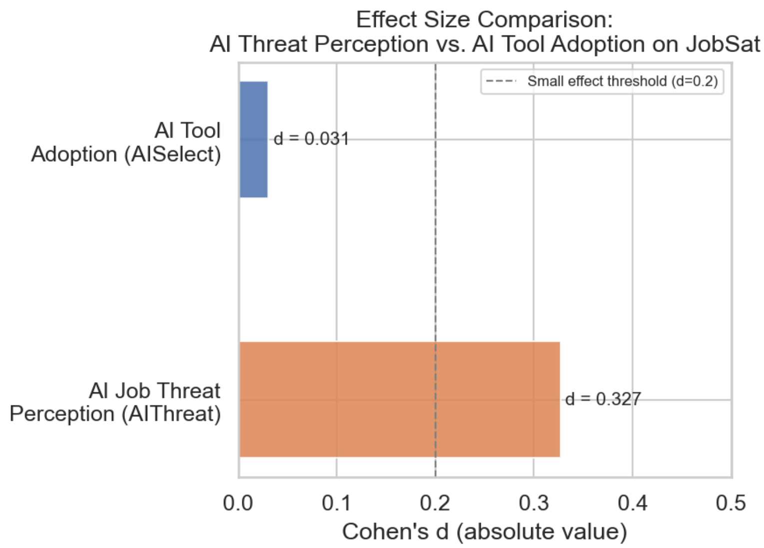

Figure 2: Effect-size comparison (Cohen's d): AIThreat vs. AISelect. Dashed line marks the conventional small-effect threshold ($d = 0.20$).

4 Discussion and Conclusion

4.1 Interpretation

The central finding is that perceiving AI as a job threat is associated with lower job satisfaction among software developers (Cohen's $d = 0.33$), while currently using AI tools is not ($d = 0.03$). This pattern is counter-intuitive in the context of the dominant discourse, which frames AI adoption as a driver of productivity gains and professional fulfilment. The data suggest, instead, that the *psychological stance* towards AI matters far more than its *behavioural adoption*.

This result is broadly consistent with Sadeghi [2024], who argues that AI's impact on employee well-being is mediated by subjective perceptions of fairness and job security rather than by the efficiency gains the technology delivers. It also aligns with Feng et al. [2025], who found that AI-related workplace anxiety reduces life satisfaction through negative emotional states in a cross-sectional sample of service workers. Our study extends this evidence to the domain of software development using a considerably larger sample.

The null effect of AISelect echoes the observation by Giuntella et al. [2025] that objective AI exposure (operationalised at the occupational level) does not reduce life satisfaction. Taken together, these findings suggest a two-tier picture: at both the macro level (occupational exposure, Giuntella et al. 2025) and the individual behavioural level (tool adoption, this study), AI's relationship with well-being is near-zero. Subjective threat perception constitutes the critical pathway.

That the threat effect persists fully among current AI users (d = 0.33 vs. 0.33 overall) is notable: using AI tools does not protect developers from the negative effects of perceiving those same tools as a job threat. This points to a form of *cognitive dissonance*—continued tool adoption alongside genuine fear of displacement—that warrants attention in both research and organisational practice. The systematic review by Soulami et al. [2024] identifies workplace autonomy and psychological health as the key mechanisms through which AI affects employees; our findings suggest that for software developers, perceived job insecurity is the dominant active mechanism.

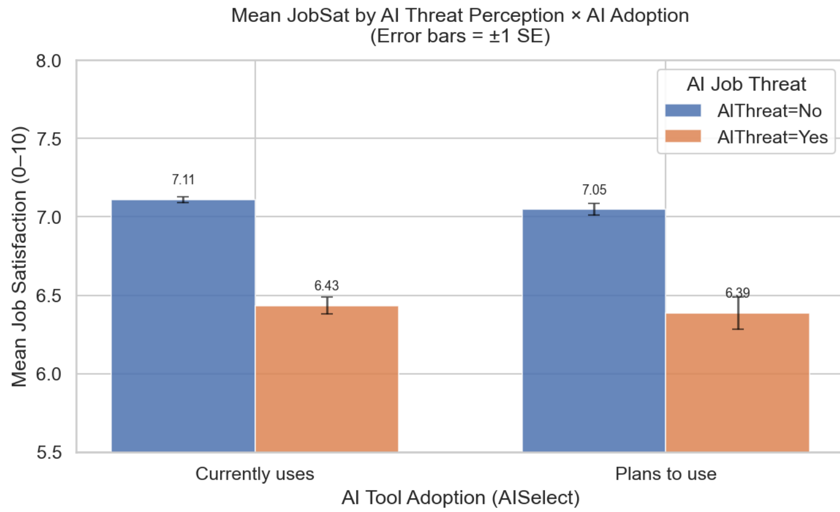

Figure 3: Mean JobSat (±1 SE) by AIThreat × AISelect. The AIThreat gap is consistent across AI adoption groups, confirming no meaningful interaction.

4.2 Practical Significance

The effect is statistically robust but modest in magnitude: Cohen's $d = 0.33$ corresponds to a mean difference of 0.67 points on a 0–10 scale, and the full model explains only 3.3 % of variance in job satisfaction. This is consistent with the cross-sectional, self-report nature of the data and the fact that job satisfaction is determined by many factors beyond AI perceptions. The small f^2 value (0.011) should be clearly communicated: AIThreat is a statistically detectable but not dominant predictor of job satisfaction.

4.3 Limitations

First, the data are cross-sectional; no causal inference is possible. Reverse causation (already dissatisfied developers may be more likely to perceive AI as threatening) cannot be ruled out. Second, JobSat was missing for 55.5 % of the full sample, restricting the analytic sample to respondents with an employment context. If dissatisfied developers systematically avoided the JobSat question, results may underestimate the threat effect. Third, AIThreat is a single binary item, providing a coarse operationalisation of what is likely a multi-dimensional attitude. Fourth, the survey captures perceived threat at one point in time during a period of rapid AI adoption; results may not generalise across survey years.

4.4 Conclusion

Against the backdrop of widespread claims that AI tool adoption drives developer satisfaction, this study finds that adoption behaviour is effectively unrelated to job satisfaction ($d = 0.03$). Perceived AI job threat, by contrast, is a consistent negative predictor ($d = 0.33$, $\beta = -0.65$, $p < 10^{-44}$) that persists across AI adoption groups and after covariate adjustment. Organisations seeking to improve developer well-being in the age of AI may gain more from addressing threat perceptions—through transparent communication, upskilling programmes, and involvement in AI rollout decisions—than from accelerating adoption rates alone.

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